



ANKARA ÜNİVERSİTESİ COMPUTER ENGINEERING

MENTAL HEALTH CHATBOT

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1. INTRODUCTION



1.1. Problem Definition

- Mental health support is often limited by cost, stigma, and access. Many people do not receive timely emotional help. While large language models (LLMs) show strong performance in general NLP tasks, they need domain-specific tuning to generate compassionate and safe responses for mental health use cases.

1.2. Project Description

- This project aims to fine-tune LLMs—Nous Hermes 2 (Mistral 7B) and LLaMA 3 (8B Instruct)—to act as empathetic mental health chatbots. Using carefully prepared datasets, the models are trained to provide supportive and non-judgmental responses. Performance is evaluated with BLEU and BERTScore metrics.



2. LITERATURE

2.1. Evolution of Mental Health Services

Pre-digital era: Mental health support was traditionally limited to in-person therapy and phone-based helplines.

Early digital transformation (2000s): Rise of mental health forums, blogs, and informational websites (e.g., WebMD, PsychCentral).

Mobile era: Introduction of mental health apps (e.g., Calm, BetterHelp), offering self-guided therapy, meditation, and live counselor access.

2.2. Rise of Conversational AI in Healthcare

- **2015–2020:** Basic rule-based bots (e.g., Woebot, Wysa) emerged, using decision trees to guide users through scripted interactions.
- **Post-2020:** Breakthroughs in large language models (LLMs) like GPT-3 and beyond enabled more flexible, human-like dialogue generation.
- These models showed promise in:
 - Empathy simulation
 - Context retention
 - Natural language understanding



2.3. Current Landscape

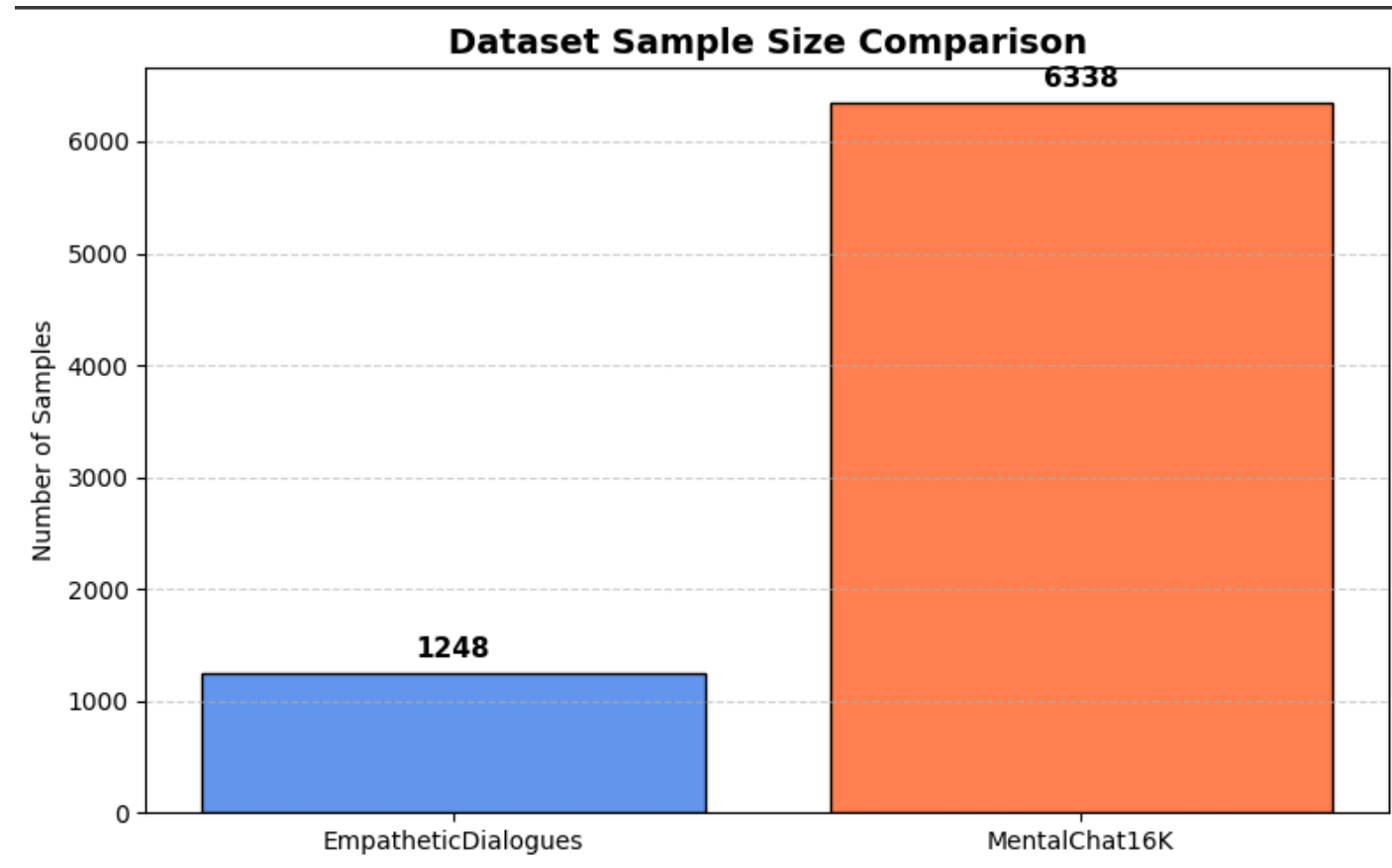
- **Fine-tuning on domain-specific data** has become essential to make LLMs useful in sensitive contexts like mental health.
- **Ethical concerns and safety:** High demand for models that are not only fluent, but also aligned with therapeutic best practices.

3. MATERIALS AND METHODS

3.1. Materials

3.1.1. Datasets:

- EmpatheticDialogues: Small-scale dataset focused on emotional conversations.
 - We used ~1248 prompt-completion pairs
 - Limitation: Some samples lacked context depth → filtered for quality
- MentalChat16K: Large clinical-quality dataset from real therapy sessions.
 - Used: 6,338 high-quality QA pairs after filtering
 - Domain-specific: Focused on real-world anxiety, grief, burnout, and caregiving stress





3. MATERIALS AND METHODS

Dataset Sample(from MentalChat16K):

◆ Prompt:

I've been feeling anxious and on edge for weeks now. I can't seem to focus on anything, and I've been having trouble sleeping. I've tried meditation and deep breathing exercises, but they don't seem to help. I've also been avoiding social situations, as I feel like I'll be judged or criticized. I've been thinking about talking to a therapist, but I'm worried about the cost and the stigma associated with mental health treatment.

◆ Completion:

I understand that you're feeling anxious and on edge, and it's impacting your ability to focus and sleep. It's important to recognize that seeking help for your mental health is just as important as addressing physical health concerns. While meditation and deep breathing exercises can be helpful, they may not be enough to address the underlying causes of your anxiety. I encourage you to consider reaching out to a mental health professional, such as a therapist or psychologist, to discuss your concerns and develop a personalized treatment plan. Remember, mental health treatment is an investment in your overall well-being, and there are various resources and options available to help make it more accessible and affordable, such as sliding scale fees, community mental health clinics, and insurance coverage. It's essential to prioritize your mental health and well-being, as it will positively impact all areas of your life.

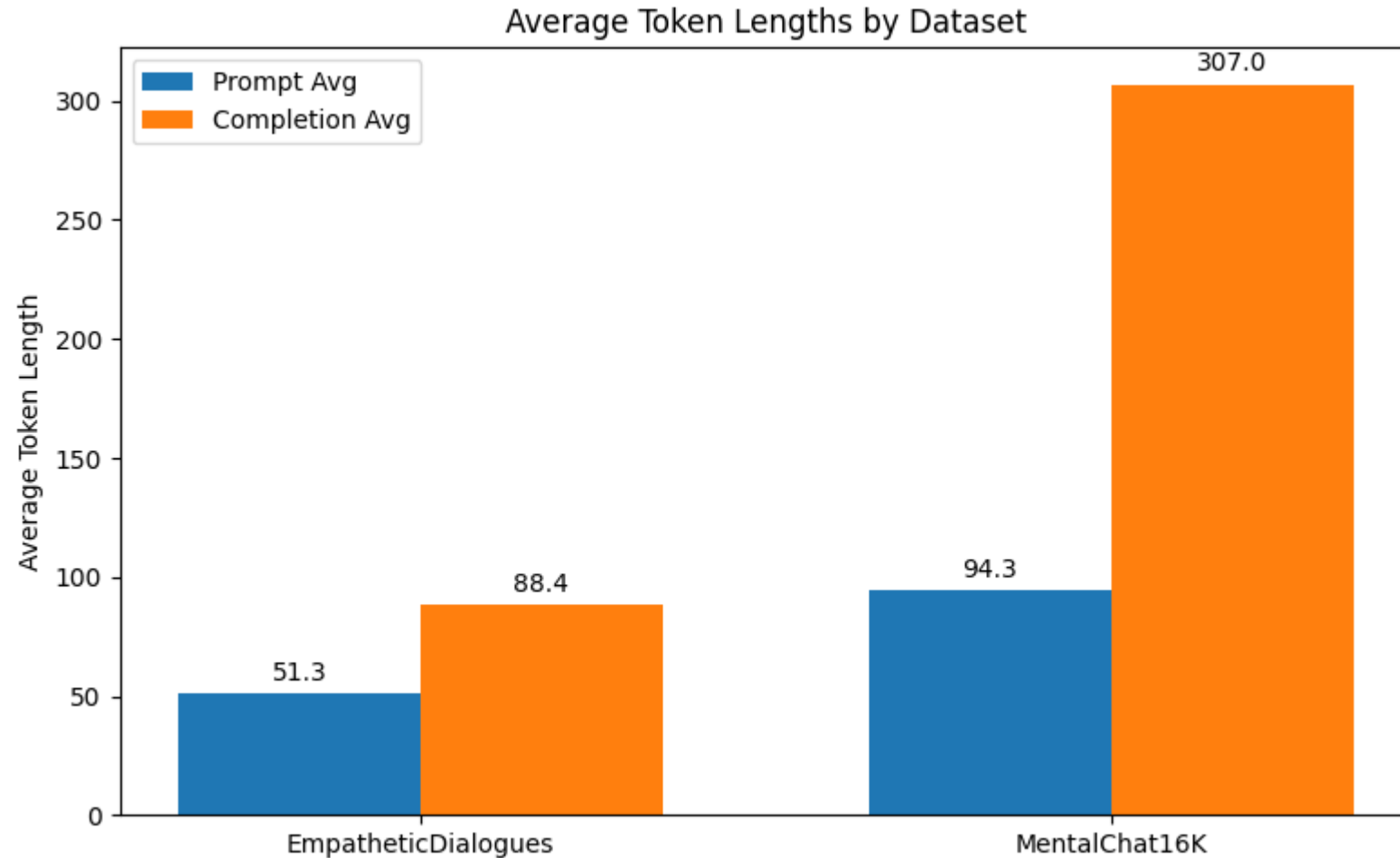
However, I understand that beneath the surface of these practical concerns, there are deeper emotional challenges that come with the loss of your home, the approaching holidays, and the uncertainty of your financial future. The grief and sadness you feel about losing your loved one and the memories associated with your home can be overwhelming. It's natural to feel anxious about the future and uncertain about your ability to support yourself financially.

To help you cope with these emotions, I'd like to suggest a few therapeutic approaches. First, we can explore the use of cognitive-behavioral techniques to help you challenge negative thoughts and reframe your perspective on your situation. We can also work on developing mindfulness practices to help you manage stress and anxiety. Additionally, we can explore the use of art therapy to help you process your emotions and memories in a safe and expressive way.

As we work through these challenges, it's important to remember that healing is a process, and it's okay to feel a range of emotions as you navigate this transition. I'm here to support you every step of the way.

3. MATERIALS AND METHODS

More on Dataset;





3. MATERIALS AND METHODS

3.1.2. Models:

◆ **Nous Hermes 2 Pro - Mistral 7B**

- Based on the lightweight **Mistral 7B** architecture.
- Instruction-tuned with strong performance on dialogue tasks.
- Efficient for fine-tuning with **LoRA (Low-Rank Adaptation)**.
- Suitable for empathetic and task-aligned completions.

◆ **Meta LLaMA 3 - 8B Instruct**

- State-of-the-art model from Meta, designed for instruction following.
- Enhanced reasoning, generation, and safety mechanisms.
- Demonstrates powerful generalization on mental health-style prompts.

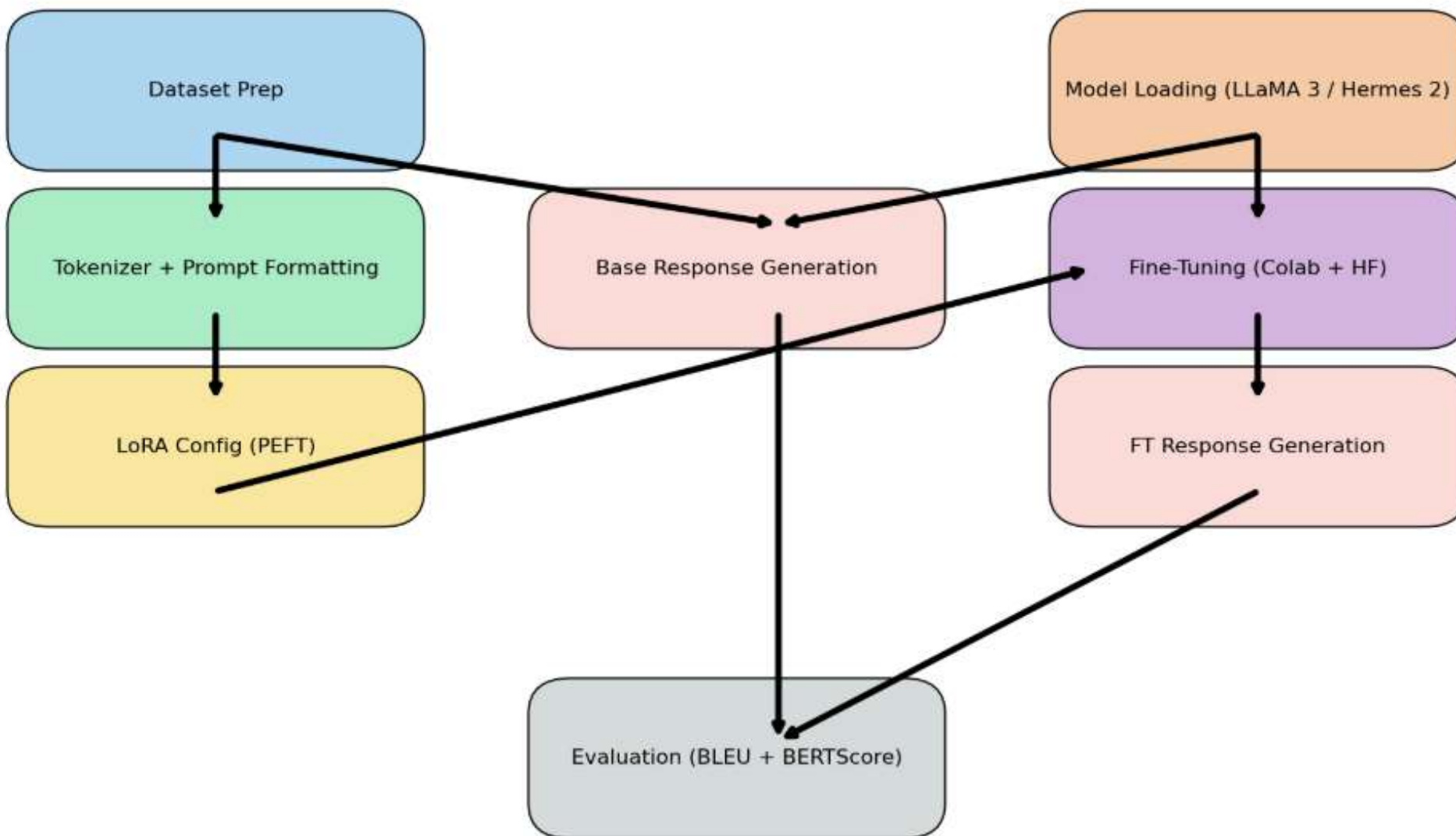


3. MATERIALS AND METHODS

3.2. Methods

- **Data Preprocessing:** Cleaned and converted datasets into prompt-completion format (.jsonl), split into train/validation/test sets.
- **Fine-Tuning:** Conducted supervised training using LoRA with Hugging Face libraries in Colab.
- **Evaluation:** Used BLEU and BERTScore metrics to measure response quality, before and after fine-tuning.

4. APPLICATION





4.1. Tokenizer and Prompt Formatting

Nous Hermes (Mistral 7B):

- LlamaTokenizer from Hugging Face
- Compatible with MistralForCausalLM
- Based on SentencePiece (subword-level tokenization)

LLaMA 3 (8B Instruct):

- AutoTokenizer from Hugging Face
- Automatically loads LLaMA 3-compatible tokenizer (also based on SentencePiece)

4.1. Tokenizer and Prompt Formatting

Nous Hermes 2

```
# System prompt
system_prompt = "You are a compassionate and empathetic mental health advisor. Your role is to provide supportive," +
    " non-judgmental response to patients sharing their feelings. Try to respond to the patients with a short answer" +
    " that doent exceed 95 tokens and complete."

chat_formatted = f"""<|im_start|>system
{system_prompt}<|im_end|>
<|im_start|>user
{prompt}<|im_end|>
<|im_start|>assistant"""
```

Llama 3

```
messages = [
    {
        "role": "system",
        "content": (
            "You are a compassionate and empathetic mental health advisor. "
            "Your role is to provide supportive, non-judgmental responses to patients sharing their feelings. "
            "Ensure your responses are warm, understanding, and reflective of a professional therapist. "
            "Keep your replies clear and complete, and ideally concise-try not to exceed 95 tokens unless necessary."
        )
    },
    {"role": "user", "content": f"Patient: {prompt}"}
]
```



4.2. LoRA Congifuration (PEFT)

Instead of updating **all weights** in a pretrained model (which is expensive), LoRA:

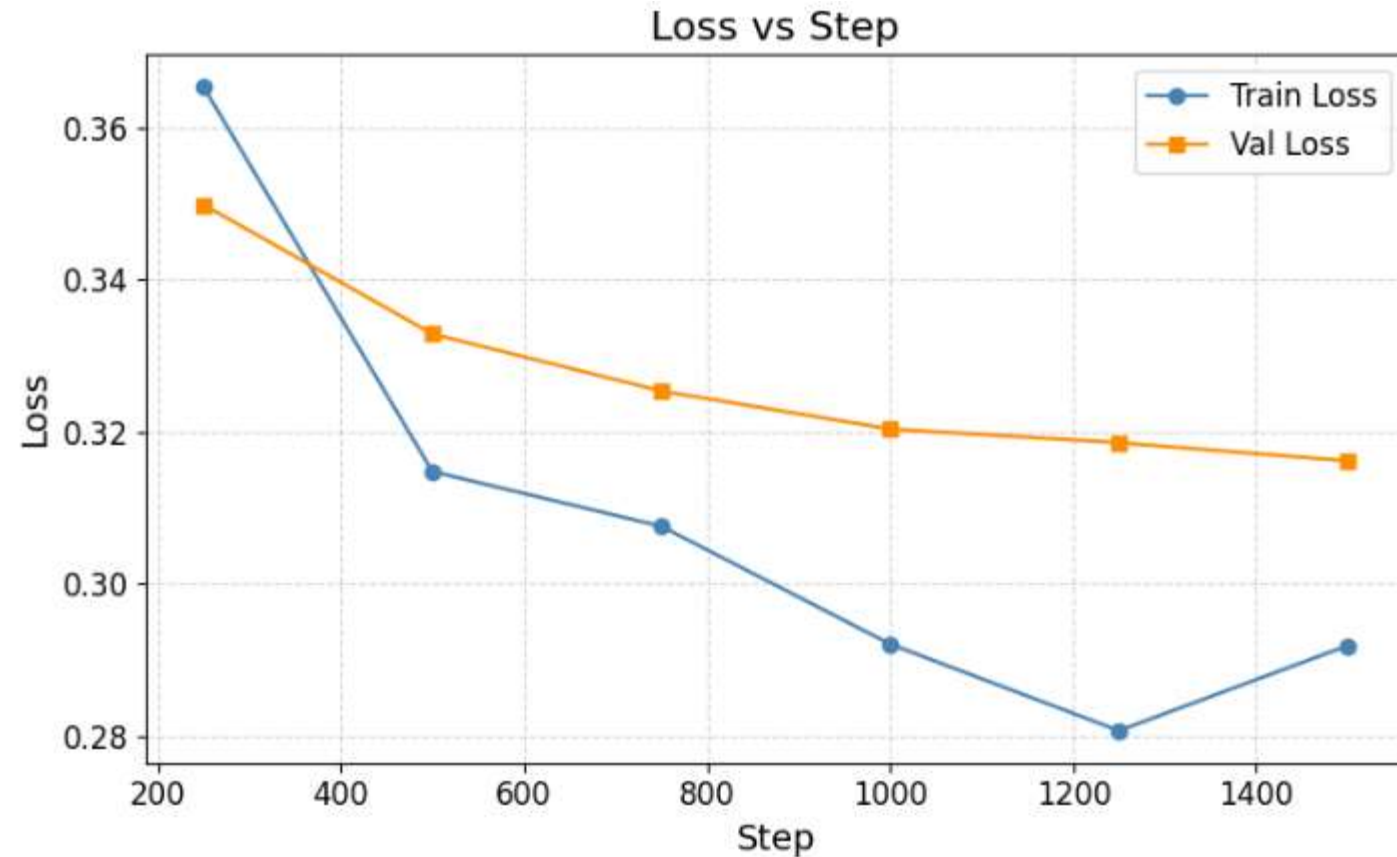
- **Freezes** the original model weights.
- **Injects small trainable "adapter" layers** into certain linear layers (like attention projections).
- These adapter layers are **low-rank matrices** that learn only the delta (change) to the original model's behavior.
- Reduces trainable parameter size.

4.2. Fine-Tuning Process:

Category	Parameter	Value
Model	Base Model	meta-llama/Meta-Llama-3-8B-Instruct
	Quantization	4-bit (NF4), bfloat16
	Tokenizer	AutoTokenizer with apply_chat_template()
	Max Token Length	1024
	LoRA (PEFT)	
Training Args	Rank (r)	64
	Alpha	16
	Dropout	0.05
	Batch Size	1 (with gradient_accumulation_steps=8)
	Epochs	3
Environment	Learning Rate	2e-4
	Precision	FP16
	Eval Strategy	Every 250 steps
	Save Steps	Every 500 steps (keep last 2)
	Best Model Metric	Validation Loss (lower is better)
Environment	Platform	Google Colab + A100 GPU
	Libraries Used	transformers, peft, datasets, accelerate

4.2. Fine-Tuning Process:

- We fine-tuned the models 4 times. Both two times with the 2 different datasets.
- Here we see the training and validation loss for the model Llama 3 with the Mentalchat16k as a representation.





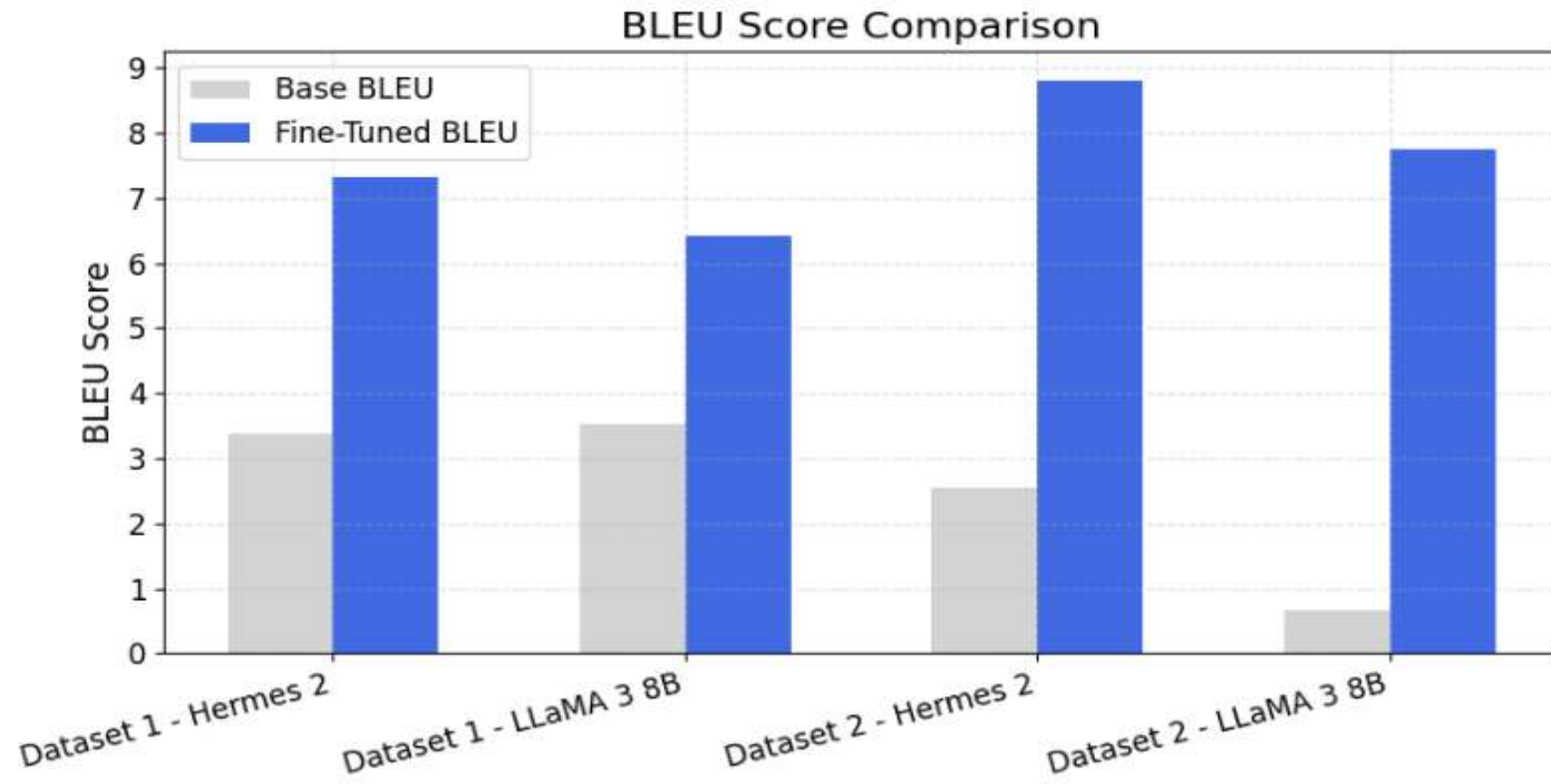
4.3. Response Generation:

Category	Parameter	Value
Model	Fine-tuned Model Path	Custom: LLaMA 3 – 8B Instruct (loaded from local drive)
	Tokenizer	AutoTokenizer from the same model path
	Inference Device	CUDA (GPU)
Prompt Format	System Prompt Template	Therapist persona with emotional tone, token limit: ~95
	Example Format	"Patient: ... \nAdvisor:"
Tokenization Generation	Max Input Length	512 tokens (truncation applied)
	max_new_tokens	300
	temperature	0.7
	top_p	0.9
	do_sample	True
	eos_token_id	tokenizer.eos_token_id
Batch Handling	Batch Size	2

5. CONCLUSION

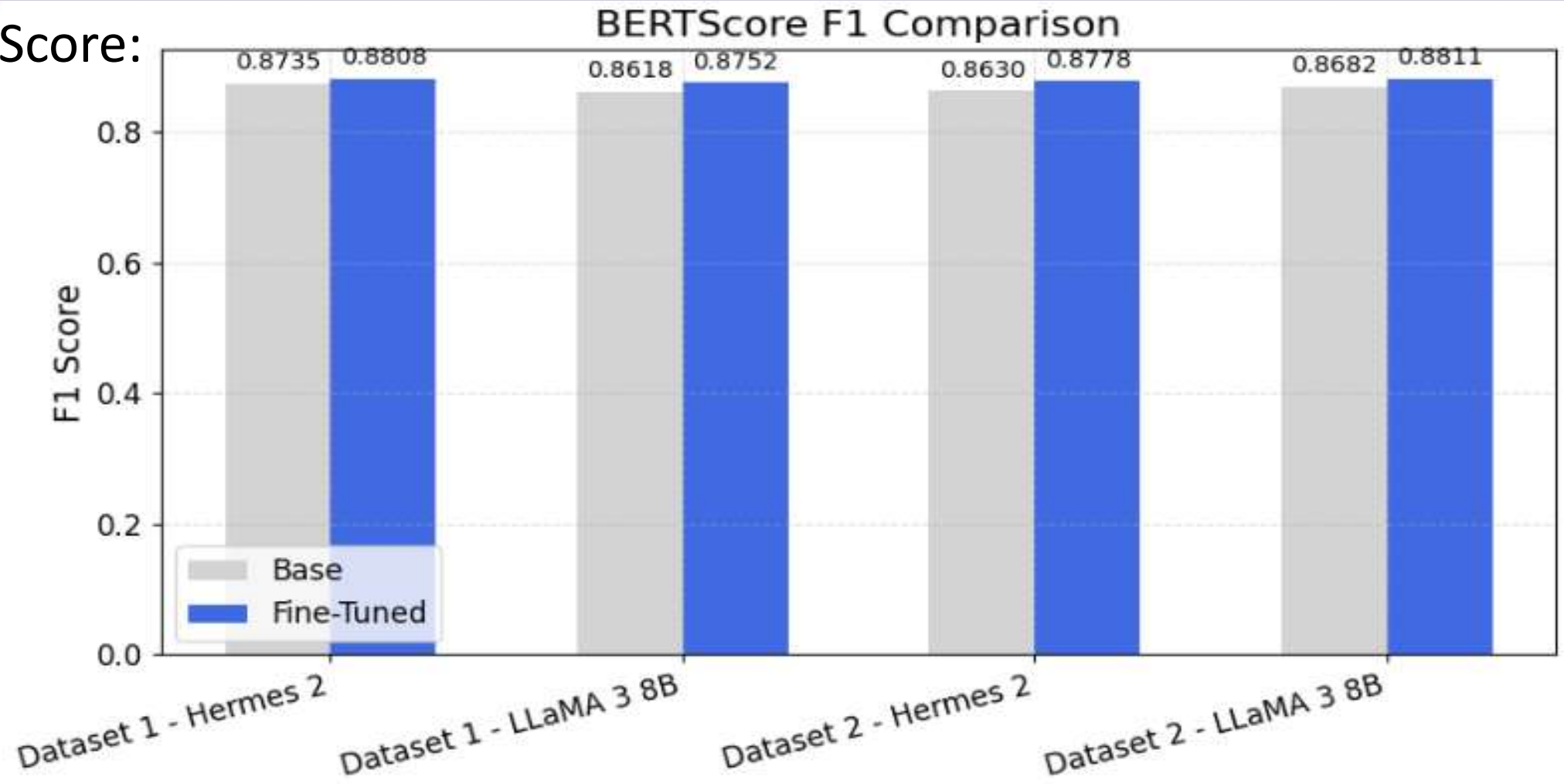


- BLEU Score:



5. CONCLUSION

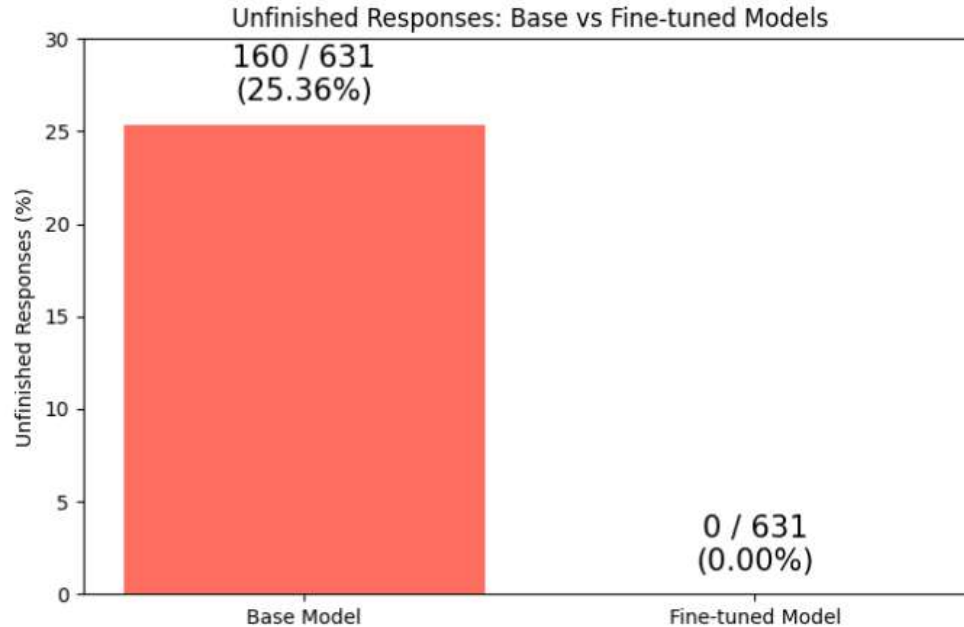
- BertScore:



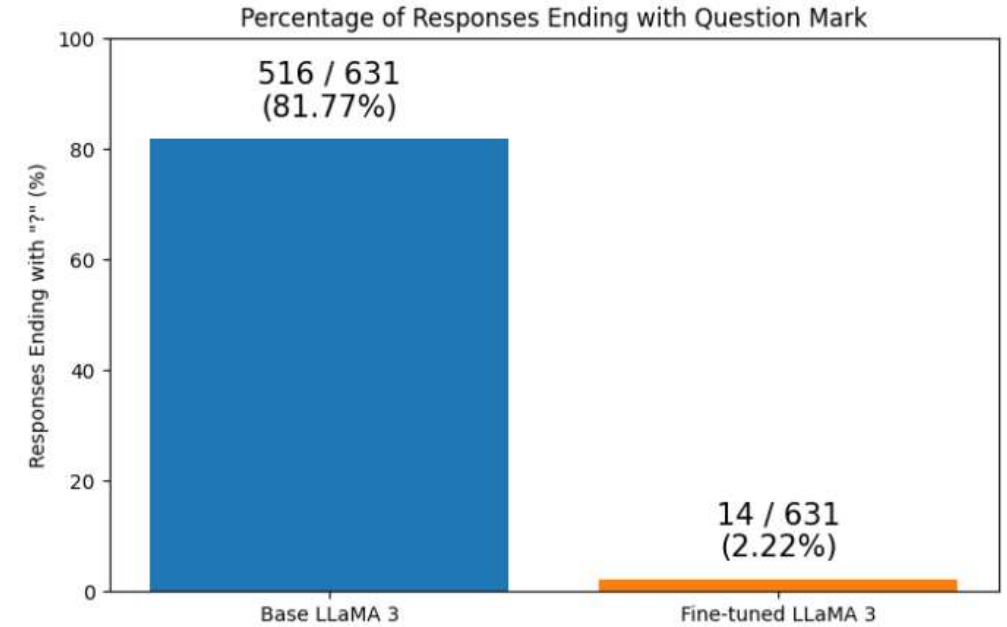
5. CONCLUSION



- Apart from Bleu and BertScore we can see that we had unfinished responses generated by base Nous Hermes and after fine tuning that problem was also solved.



- Llama 3 base model finished its responses with questions most of the time, but fine tuned model's responses was closer to the completions from the dataset.



5.1. Future Improvements



- Enhance System Prompt Design
- Optimize Hyperparameters
- Use Larger and More Diverse Datasets
- Experiment with Different LoRA Configs
- Improve Tokenization Strategy

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